MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7402

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: High vacuum clipper diode or rectifier

See figure 1

Mounting --- Vertical position, base down

Weight --- 10 ounces nominal

ABSOLUTE-RATINGS:

Parameter:		Ef	epx	ib	Ib	Ieff	Pp
Unit:		Vac	kv	a	mAdc	Aac	W
Clipper	Max:	12.1	33	50	----	1.25	450
	Min:	10.9	--	---	----	----	---
Rectifier	Max:	12.1	33	2.7	860	----	450
	Min:	10.9	--	---	----	----	---
TEST CONDITIONS:		11.5	--	---	----	----	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	14.75	16.25	Aac
1231	Peak emission	eb = 8.0 kv	0.65	II	is	30	---	a
1256	Electrode current (anode)	Eb = 330 Vdc	0.65	II	Ib	860	---	mAdc
1353	Operation of rectifiers	epx = 33 kv; ib = 2.7 a Ib = 860 mAdc (see note 1)	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low frequency vibration	No voltages	6.5	S3	---	---	---	---
1041	Shock	Hammer angle 25°	---	---	---	---	---	---
----	Operation	epx = 33 kv tp = 1 µsec; Zo = 53 ohms; Rs = 300 ohms(min); Prr = 340 Rl = 0 (See notes 1 and 2)	---	---	ib	50	---	a
	Quality conformance inspection, part 3							
----	Life test provisions	Group D Operation of rectifiers	---	---	t	500	---	hrs
----	Life test end point:							
1231	Peak emission		---	---	is	25	---	a

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NOTES:

1. The tube shall operate for 2 minutes without arcing or signs of gas during a test interval not to exceed 5 minutes.

2. The tube shall be operated as a clipper diode in a thyratron modulator.

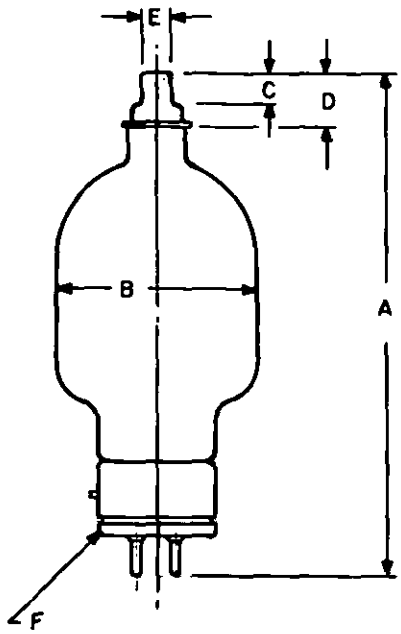
Custodian:
Air Force - 80

Preparing activity:
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

(Project 5960-F866)

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2				
A	9.250	9.750	235.15	247.65
B	–	3.625	–	92.07
C	.594	.656	15.09	16.66
QUALITY CONFORMANCE INSP. PART 3 (PERIODIC CK)				
E	.559	.573	14.20	14.60
F	BASE A-4-18			
REFERENCE DIMENSIONS				
D	1.00		25.40	

FIGURE 1. Outline drawing of electron tube 7402

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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